Why the Proton Is Irreducibly Tripartite: the color-singlet obstruction to bipartite qudit merges

Tessera — cobordism subsystem (#382 / #396)

Abstract

We attempted to build a proton by *composing bipartite merges*: relax a cobordism whose boundary is two color states ψ_A, ψ_B, read the emergent result ψ_{AB}, then feed it as a boundary input of a second merge with a third state ψ_C. The emergent result never reaches the color singlet — it stays color-charged ($|\sigma| \approx 0.67$) at every step. This note shows that this is *not* a numerical or topological defect of the construction but a representation-theoretic theorem: the color singlet of $SU(3)$ is the three-index alternating invariant ε_{ijk}, which is *absent* from the two-fold tensor product $\mathbf{3} \otimes \mathbf{3}$ and first appears in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. A baryon singlet built from fundamentals is therefore irreducibly an $n = 3$ object; no single bipartite fusion of two color triplets can produce it. The qudit generalization makes the statement sharp: for $SU(d)$ the baryon arity equals d, so a *qubit* ($d = 2$) baryon *is* bipartite (the spin singlet ε_{ij}), while a *qutrit* color baryon ($d = 3$) is not. The simplicial resolution is the trivalent junction W_{ABC} that carries ε_{ijk} directly.

1 The question

The merge primitive is a Lorentzian Regge cobordism W with prescribed boundary ∂W: we pin the boundary color states, relax the interior edge lengths to a stationary point of the dual Regge action subject to a realizability constraint, and read the emergent result off the relaxed geometry. Composition is by *cobordism*, not welding: the emergent result-state of one merge is the boundary input of the next.

Concretely we merged three color-neutral $q\bar{q}$ pairs $A = (1, -1, 0)$, $B = (1, 0, -1)$, $C = (0, 1, -1)$ pairwise:

$$\text{merge}(A, B) \to AB, \qquad \text{merge}(AB, C) \to ABC.$$

With the exact period read-out (`residualForPeriods` for the pinned inputs, `cyclePeriods` for the emergent result block) we measure the emergent color charge $\sigma(\psi) = \sum_i \psi_i$:

$$|\sigma_{AB}| \approx 0.70, \qquad |\sigma_{ABC}| \approx 0.67, \qquad \text{target singlet } |\sigma| = 0.$$

The result is color-charged at every step. *Is this a bug or a theorem?* It is a theorem.

2 Color, invariant tensors, and singlets

A quark transforms in the fundamental $\mathbf{3}$ of $SU(3)_{\text{color}}$, an antiquark in $\bar{\mathbf{3}}$. A physical (observable) hadron must be *colorless*: a singlet, i.e. an invariant under the color group. Building hadrons is therefore the problem of constructing $SU(3)$-invariant tensors out of copies of $\mathbf{3}$ and $\bar{\mathbf{3}}$.

Theorem 1 (First fundamental theorem of invariant theory for $SL(3)$)**.** *Every $SU(3)$-invariant tensor in copies of $\mathbf{3}$ and $\bar{\mathbf{3}}$ is a polynomial in the two elementary invariants*

$$\delta^i{}_j \quad (\textit{the contraction } \mathbf{3}\otimes\bar{\mathbf{3}}\to\mathbf{1}), \qquad \varepsilon_{ijk},\ \varepsilon^{ijk} \quad (\textit{the alternating } \Lambda^3).$$

The two invariants are exactly the two ways to make a colorless object:

- $\delta^i{}_j$ pairs a quark with an antiquark: the **meson** $\bar{q}_i\, q^i$. It is *bilinear* — a colorless meson is a two-body (bipartite) object.
- ε_{ijk} contracts *three* quarks: the **baryon** $\varepsilon_{ijk}\, q^i q^j q^k$ (the proton). It is *trilinear* and totally antisymmetric.

The decisive fact is what is *missing*: there is no invariant bilinear in two fundamentals.

Proposition 1 (No singlet at two bodies)**.**

$$\mathbf{3}\otimes\mathbf{3} \;=\; \mathbf{6}\,\oplus\,\bar{\mathbf{3}}, \qquad \textit{and neither summand is the trivial representation } \mathbf{1}.$$

Hence the two-quark Hilbert space $\mathbf{3}\otimes\mathbf{3}$ contains no *color singlet. Equivalently, the multiplicity of $\mathbf{1}$ in $\mathbf{3}^{\otimes 2}$ is* 0*.*

The singlet first appears one body later:

Proposition 2 (The singlet is born at three bodies)**.**

$$\mathbf{3}^{\otimes 2} = \mathbf{6}\oplus\bar{\mathbf{3}} \quad (\dim 9 = 6+3), \qquad \mathbf{3}^{\otimes 3} = \mathbf{10}\oplus\mathbf{8}\oplus\mathbf{8}\oplus\mathbf{1} \quad (\dim 27 = 10+8+8+1).$$

The trivial representation occurs with multiplicity $0, 0, 1$ *in* $\mathbf{3}^{\otimes 1}, \mathbf{3}^{\otimes 2}, \mathbf{3}^{\otimes 3}$*; the unique copy in $\mathbf{3}^{\otimes 3}$ is spanned by ε_{ijk}.*

Proposition 1 is the heart of the matter: *the proton state is not an element of any two-body color Hilbert space.* A bipartite fusion $V_A\otimes V_B\to V_R$ of two color triplets can only land in $\mathbf{6}\oplus\bar{\mathbf{3}}$; its output spectrum simply does not contain the singlet to which we are trying to relax. Compare the colorless meson: it lives in $\mathbf{3}\otimes\bar{\mathbf{3}} = \mathbf{1}\oplus\mathbf{8}$, which *does* contain $\mathbf{1}$ — which is exactly why a $q\bar{q}$ pair (our carriable input) is a legitimate two-body colorless object, while a proton is not.

3 The associativity subtlety: why a *sequence* of merges still fails

One might object that the singlet *is* reachable pairwise through associativity:

$$(\mathbf{3}\otimes\mathbf{3})\otimes\mathbf{3} = (\mathbf{6}\oplus\bar{\mathbf{3}})\otimes\mathbf{3} = \underbrace{(\mathbf{6}\otimes\mathbf{3})}_{\mathbf{10}\oplus\mathbf{8}} \;\oplus\; \underbrace{(\bar{\mathbf{3}}\otimes\mathbf{3})}_{\mathbf{1}\oplus\mathbf{8}}.$$

The singlet appears *only* in the $\bar{\mathbf{3}}\otimes\mathbf{3}$ channel: one must first project the diquark onto its *antisymmetric* part $\bar{\mathbf{3}} = \varepsilon_{ijk}q^jq^k$ and discard the symmetric sextet $\mathbf{6}$. A bipartite merge that *projects onto the antisymmetric channel* could then complete the singlet with the third quark.

But the emergent merge does no such projection. Relaxing the cobordism produces a stationary geometry carrying the *generic* product $\mathbf{6}\oplus\bar{\mathbf{3}}$; the symmetric $\mathbf{6}$ component is not annihilated, and it contaminates the second merge — the result stays colored ($|\sigma_{AB}|\approx 0.70$). To kill the $\mathbf{6}$ one must impose the *simultaneous* antisymmetry of all three color indices, ε_{ijk}, at a *single* vertex. That object is a trivalent junction carrying the alternating (sign) representation — not any composition of generic pairwise merges.

Remark 1. This is the precise sense in which the proton is "irreducibly tripartite": the defining invariant $\varepsilon_{ijk}\in\Lambda^3(\mathbb{C}^3)^*$ is a totally antisymmetric *three*-form. It does not factor as a composite of bilinear invariants except by first manufacturing the antisymmetric diquark $\bar{\mathbf{3}}$ — and that projection is itself the statement that the three indices must meet antisymmetrically. There is no associative bracketing that turns a *generic* two-body fusion into a singlet.

4 The qudit generalization (the sharp answer)

The argument is not special to $SU(3)$; it pins the baryon *arity* exactly. For $SU(d)$ acting on a qudit (the fundamental $\mathbb{C}^d$), the invariant tensors are again generated by $\delta^i{}_j$ and the Levi-Civita symbol $\varepsilon_{i_1\cdots i_d}$, now with d *indices*.

Corollary 1 (Baryon arity $= d$)**.** *A singlet built from fundamentals of $SU(d)$ requires exactly d of them: $\varepsilon_{i_1\cdots i_d} q^{i_1}\cdots q^{i_d}$. The multiplicity of $\mathbf{1}$ in $\mathbf{d}^{\otimes n}$ is 0 for $0 < n < d$ and first becomes nonzero at $n = d$.*

This is the direct answer to "why can't bipartite qubit/qudit interactions reach the proton":

- **Qubit,** $d = 2$**.** ε_{ij} has *two* indices: $\mathbf{2}\otimes\mathbf{2} = \mathbf{3}\oplus\mathbf{1}$ already contains a singlet (the spin-0 state $\frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$). A "baryon" of qubits *is* bipartite. Bipartite merges suffice here — which is exactly why qubit intuition misleads.
- **Qutrit / color,** $d = 3$**.** ε_{ijk} has *three* indices: $\mathbf{3}\otimes\mathbf{3} = \mathbf{6}\oplus\bar{\mathbf{3}}$ has no singlet. The proton is a qutrit baryon and is irreducibly *tripartite*.

The proton's three-quark structure is, in one line, the statement $d = 3$: the color singlet is the d-fold alternating invariant, and $d = 3$ for color.

5 The simplicial model and the measurement

In the cobordism subsystem the color register is the homology of the holed register surface: $b_1 = 2$ on the $\sigma = 0$ hyperplane, carrying the *standard* (2-dimensional) representation of $S_3 = \mathrm{Weyl}(SU(3))$ permuting the three colors, with

$$\mathbb{C}^3 = \underbrace{\mathbf{1}}_{\sigma\neq 0 \text{ direction}} \oplus \underbrace{V_{\text{std}}}_{\sigma=0 \text{ register}} .$$

Confinement is the topological statement that the $\sigma \neq 0$ (trivial-rep) direction is *not carried* by the register: a net color charge has no realizable harmonic representative, while a $\sigma = 0$ configuration does. The proton singlet is represented by the definite-triality state

$$\psi_{\text{proton}} = (1, \omega, \omega^2), \qquad \omega = e^{2\pi i/3}, \qquad 1 + \omega + \omega^2 = 0,$$

the $\mathbb{Z}_3 \subset S_3$ Fourier mode: it is the discrete shadow of the alternating invariant ε_{ijk} (an eigenvector of the cyclic color rotation, i.e. a state of definite triality, lying in the $\sigma = 0$ register).

A bipartite merge is a cobordism W with $\partial W = \psi_A \sqcup \psi_B$ and an emergent result block R; we read $\sigma_R = \sum_i (\psi_R)_i$ from the relaxed metric via `cyclePeriods`. Consistent with Proposition 1, the emergent two-input result does not land in the singlet direction:

$$|\sigma_{AB}| \approx 0.70, \qquad |\sigma_{ABC}| \approx 0.67 \quad \text{(sequence)},$$

against the singlet target $|\sigma| = 0$. Two control facts confirm the reading is faithful and the obstruction structural:

- The result is read with the *exact* period machinery (`residualForPeriods/cyclePeriods`), not the soft edge-loop encoding; the $\approx 10^{-2}$ neutral realizability floor is intrinsic to the shared-register staircase, not an artifact of the read-out. The colored/neutral confinement split (ratio $\sim 10^3$) is intact.
- Each merge step is a genuine valid manifold ($b_1 = 2$, every triangle in ≤ 2 tetrahedra), so the failure to reach the singlet is not a degeneration of the cobordism but a property of the two-body color content.

6 Resolution: the trivalent junction W_{ABC}

The representation theory dictates the fix. Since ε_{ijk} is a *three*-valent invariant, the proton must be realized by a cobordism with *three* boundary inputs meeting a single bulk,

$$\partial W_{ABC} = \psi_A \sqcup \psi_B \sqcup \psi_C,$$

a trivalent junction whose induced orientation on the three color cycles carries the alternating (sign) structure — the simplicial ε_{ijk}. This is the $n = 3$ object in which the singlet first occurs, built once rather than as a sequence of two-body fusions. The exact period machinery used here (`readoutHoles`/`residualForPeriods` to pin the three inputs, `cyclePeriods` to read the emergent result, `emergesResult` to pin inputs only) is exactly what that junction consumes; only the junction topology itself is new (tracked as #396).

7 Conclusion

The proton does not emerge from bipartite merges because the $SU(3)$ color singlet is the three-index alternating invariant ε_{ijk}, which is absent from $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and first appears in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$. The two-quark color Hilbert space simply does not contain the state we were relaxing toward; a sequence of generic pairwise fusions cannot manufacture the simultaneous antisymmetry of three color indices. The qudit form of the statement is exact — baryon arity $= d$, so qubits make baryons bipartitely but qutrit color does not — and our simplicial register reproduces it quantitatively: the emergent bipartite result stays color-charged ($|\sigma| \approx 0.67$) at every step. The proton is irreducibly tripartite, and its carrier is the trivalent junction W_{ABC}.